Hydrocracker Unit Second-Stage Catalyst Kinetic Model & Quench Spec

1. Proprietary Catalyst Chemistry and Operating Envelope

CONFIDENTIAL NOTICE: This document contains proprietary catalyst formulation data for the Second-Stage Hydrocracker Reactor R-901 at Mangalore Refinery and Petrochemicals Limited. Access is restricted to Plant Managers and accredited Safety Auditors with SECRET security clearance.

Catalyst Specifications:
- Active Metals: Sulfided Nickel-Molybdenum (Ni-Mo) supported on dealuminated modified Y-Zeolite carrier.
- Surface Area: 340 m2/g with tailored micropore distribution optimizing middle-distillate selectivity.
- Design Operating Pressure: Total pressure = 162.0 bar gauge | Hydrogen Partial Pressure P_H2 >= 145.0 bar.
- Liquid Hourly Space Velocity (LHSV): Nominal 1.25 hr^-1 (allowable range 1.05 to 1.45 hr^-1).
- Bed Operating Temperature Window: 370.0 deg C to 415.0 deg C (Maximum metallurgical design limit 440.0 deg C).

2. Reaction Kinetics and Exothermic Runaway Quench Control

Hydrocracking is strongly exothermic. First-order cracking rate kinetics follow the Arrhenius relationship:
Reaction Rate: k = A * exp(-E_a / (R * T))
Where Activation Energy E_a = 142.5 kJ/mol, Pre-exponential factor A = 4.82 x 10^9 s^-1, and gas constant R = 8.314 J/(mol*K).

Automated Inter-Bed Quench Gas Algorithm:
To prevent thermal hot-spot runaway in reactor R-901, cold hydrogen quench gas is dynamically injected between catalyst beds 1, 2, and 3. The automated quench controller throttles quench valves to strictly enforce:
Maximum allowable temperature rise across any single catalyst bed: Delta T_bed <= 18.0 deg C.
If Delta T_bed exceeds 22.0 deg C, automated emergency depressurization (AED) activates within 15 seconds.

HCU Reactor R-901 Yield Optimization & Compliance Mandate

3. Ultra-Low Sulfur Diesel (ULSD) Yield Optimization Matrix

Empirical yield optimization model for maximizing high-cetane BS-VI compliant diesel production:
Diesel Yield (wt%) = 58.4 + 0.32 * (Severity Index) - 0.0041 * (Severity Index)^2
Optimal Cracking Severity Index target = 68.5%.
Predicted Maximum Diesel Yield = 61.2 wt% | Heavy Naphtha Yield = 22.8 wt% | Kerosene/Jet Fuel = 11.5 wt%.
Product Sulfur Slip Specification: Finished diesel sulfur content must remain strictly below 8.0 ppm wt.
Cetane Index: Minimum 54.0 units with polycyclic aromatics content < 6.0 wt%.

4. Sovereign Air-Gapped Compliance and Clearance Safeguard

Pursuant to National Critical Information Infrastructure Protection Centre (NCIIPC) guidelines:
1. This proprietary kinetic model and catalyst composition specification is classified SECRET.
2. Querying or retrieving chunks of this specification requires verified SECRET level clearance.
3. Any unauthorized attempt by lower-clearance roles (e.g. OPERATOR or PROCESS_ENGINEER) to access these records is logged into the SHA-256 cryptographic audit ledger and triggers an immediate security alert.
4. All mathematical verifications referencing this formula must execute within the isolated local AST sandbox.